Machine-Learning Driven Identification of Pathogenic Predictors in Murine Cholelithiasis

A Comparative Study of Lithogenic Diet Response

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Outline

1. Introduction & Background
2. Computational Pipeline
3. Data & Feature Engineering
4. Model Competition & Results
5. Calibration Strategy
6. Explainability & SHAP
7. Strain-Level Error Analysis
8. Discussion & Conclusion

Project Abstract

- **385 mice** (Paigen diet), **291 retained**
- 3 engineered features: *weight_gain_pct*, *cholesterol_ratio*, *non_hdl_calc*
- Anti-leakage + missingness protocol ($> 70\%$ dropped)

Champion Model

Calibrated Logistic Regression

Recall $= 0.771$ F1 $= 0.682$

Weight Gain Paradox

weight_gain_pct is **top-3 predictor** despite **negative** marginal correlation ($r \approx -0.23$)

Introduction & Background

What is Cholelithiasis?

- **Gallstone disease** – common condition with significant morbidity
- Pathogenesis: cholesterol supersaturation $\rightarrow$ crystal nucleation $\rightarrow$ stone growth
- Modulated by: *genetic background*, *diet*, *metabolic factors*
- Murine models (Paigen diet) induce cholesterol gallstones in susceptible strains

Why ML?

Traditional methods miss **non-linear** interactions between covariates.

ML captures complex patterns and provides **calibrated risk scores**.

Study Objectives

1. Synthesise individual phenotypes with **population-level strain summaries**
2. Apply rigorous **data refinement & anti-leakage protocol**
3. Compare classifiers: **Logistic Regression, Random Forest, XGBoost**
4. Post-process with **isotonic calibration** for interpretable probabilities
5. Interpret model via **SHAP** and **strain-level error analysis**

Computational Pipeline

4-Phase Analytical Pipeline

1. **Data Harmonization**
   Individual phenotypes + Strain means merged on composite key
2. **Refinement**
   Leakage exclusion, missingness threshold, imputation
3. **Algorithmic Competition**
   LR / RF / XGBoost under 5-fold CV
4. **Interpretation**
   SHAP + Isotonic Calibration

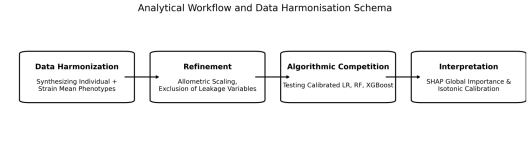

Fig 1. Analytical Workflow and Data Harmonisation Schema.

Computational Environment

Libraries & Tools:

- **Scikit-Learn** – Logistic Regression, Random Forest, `CalibratedClassifierCV`, `StratifiedKFold`, imputation
- **XGBoost** – gradient boosting with class-weighted objective
- **SHAP** – `LinearExplainer` for model interpretation

Reproducibility:

- Fixed **random seed** for all train–validation splits
- All stochastic components seeded identically
- Metrics exported to `results_plots/cv_metrics.json`

Environment

Standard workstation; optional GPU acceleration available for tree-based methods.

Data & Feature Engineering

Data Synthesis – 3 Sources into 1 Master Table

Two per-animal sources:

- Morphometric: *initbw*, *finalbw*, *liverweight*, *gbvol*
- Lipid: *totalCh*, *HDL_Ch*, *nonHDL_Ch*
- Merged on: strain × sex × age × animal ID

One strain-level summary source:

- Long → wide format (pivot)
- Prefix: `strainmean_totalCh`, `strainmean_st_pres`, ...
- Joined on strain & sex only

Result

1 unified table: individual-level + strain-level covariates per animal

Key point

Animal's own outcome is **never used** in strain means – no leakage

Dataset & Data Cleaning

Dataset:

- 385 obesity-prone mice (Paigen diet)
- 3 data sources merged on composite key
- 94 excluded (missing labels)
- **291 retained** with verified *st_pres*

Class Distribution:

- 62 Stone-positive (21.3%)
- 229 No Stone (78.7%)
- $\Rightarrow$ Substantial class imbalance
- Addressed via `class_weight='balanced'`

Feature Engineering – 3 Derived Features

1. **Weight Gain Percentage (allometric scaling)**

$$weight_gain_pct = \frac{finalbw - initbw}{initbw} \times 100$$

Normalises for strain/sex baseline variance across strains and sexes

2. **Cholesterol Ratio (atherogenic index)**

$$cholesterol_ratio = \frac{totalCh}{HDL_Ch}$$

3. **Calculated Non-HDL Cholesterol**

$$non_hdl_calc = totalCh - HDL_Ch \qquad \text{(validated against } nonHDL_Ch\text{)}$$

Anti-Leakage Protocol

Variables excluded (data leakage):

- *gs*, *nstones* – direct proxies of *st_pres*
- *st_pres* – target variable
- *animal_short_id*, *animal_id*, *comments* (identifiers or free text)

Missingness handling:

- Columns with $> 70\%$ missing $\rightarrow$ dropped (*inflammation*, *microvacs*, *macrovacs*, ...)
- Remaining: median (numeric) / most-frequent (categorical) imputation
- Same logic applied at training *and* inference time

Model Competition & Results

Model Comparison – Stratified 5-Fold CV

Model	Precision	Recall	F1
Univariate Baseline	0.29 ± 0.08	0.61 ± 0.16	0.40 ± 0.10
Calibrated LR	$\mathbf{0.615 \pm 0.04}$	$\mathbf{0.771 \pm 0.04}$	$\mathbf{0.682 \pm 0.04}$
Random Forest	0.749 ± 0.10	0.497 ± 0.12	0.589 ± 0.10
XGBoost	0.736 ± 0.07	0.597 ± 0.08	0.654 ± 0.04

Champion: Calibrated Logistic Regression

Highest **Recall** $= \mathbf{0.771} \Rightarrow$ minimises false negatives (missed stone-positive animals)

Performance Manifold

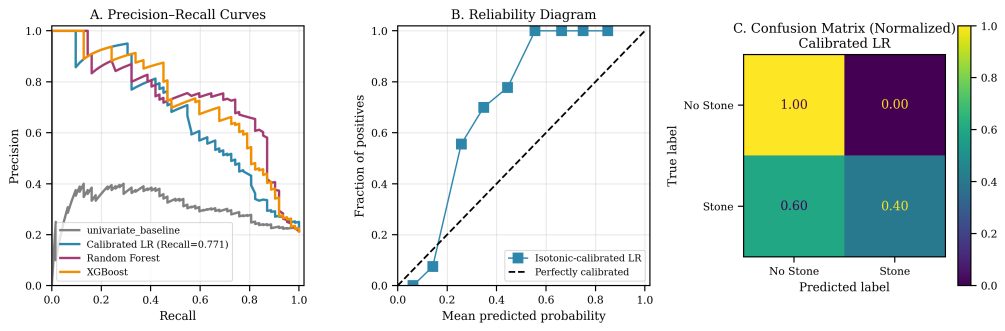

Fig 2. ***(A)*** *Precision–Recall curves for Calibrated LR, Random Forest, and XGBoost.* ***(B)*** *Reliability diagram: isotonic-calibrated LR vs. perfectly calibrated diagonal.* ***(C)*** *Normalised confusion matrix for the champion model.*

Best Model PR Curve (CV Probabilities)

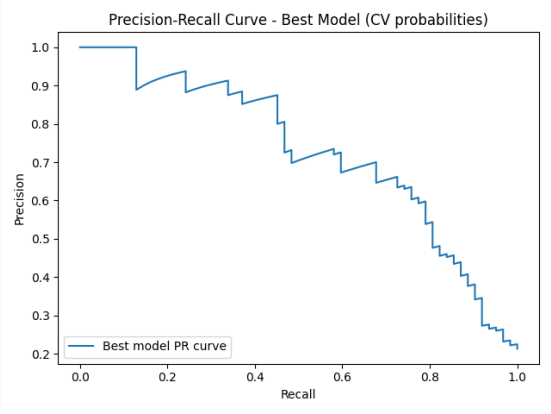

Fig 3. Precision–Recall curve of the best model (CV probabilities).

Key observations:

- High precision maintained up to Recall ≈ 0.5
- Trade-off steepens beyond Recall $= 0.7$
- Operating point chosen at **Recall = 0.771**
- Class imbalance (21% positive) – high-recall model maintains reasonable precision

Confusion Matrix – Best Model (CV Predictions)

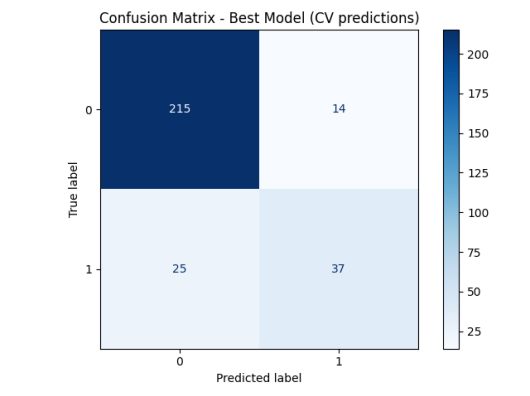

Fig 4. Confusion matrix – best model (CV predictions).

Reading the matrix:

- **TN = 215** – correctly identified No-Stone
- **FP = 14** – No-Stone wrongly flagged
- **FN = 25** – Stone-positive missed
- **TP = 37** – correctly detected Stone

Priority

Minimising **FN (25)** is critical – these are missed gallstone cases. Model correctly captures **37/62** Stone-positive animals.

Calibration Strategy

Isotonic Calibration – Reliability Diagram

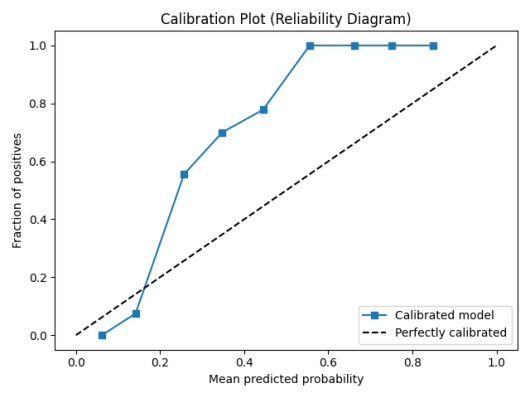

Fig 5. Calibration plot (Reliability Diagram).

What this shows:

- Calibrated model (blue) vs. perfect diagonal (dashed)
- Low probabilities (< 0.2): slightly under-predicts positives
- Mid-range (0.3–0.5): good alignment with observed rates
- High probabilities (> 0.55): fraction of positives reaches 1.0

Calibration Metrics

	Uncalib.	**Calibrated**
Brier	0.094	**0.075**
ECE	0.120	**0.099**

Explainability & SHAP

SHAP Feature Importance – Logistic Regression

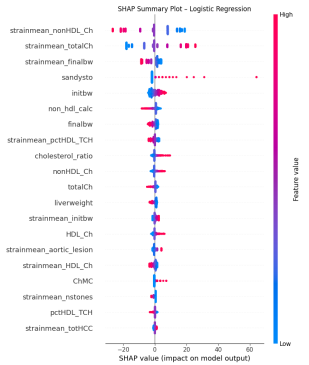

Fig 6. SHAP Summary Plot – Logistic Regression (top 20 features).

Top drivers of stone risk:

- **strainmean_nonHDL_Ch** – strain-level non-HDL cholesterol
- **strainmean_totalCh** – strain-level total cholesterol
- **sandysto** – histopathological score
- **initbw** – initial body weight
- **non_hdl_calc** – engineered feature

Pink (high value) → pushes toward Stone
Blue (low value) → pushes toward No Stone

SHAP Analysis & Weight Gain Paradox

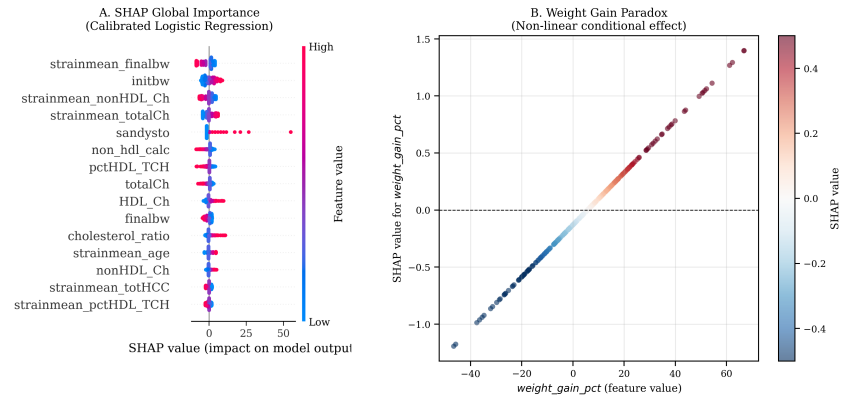

Fig 7. ***(A)*** *Global SHAP importance (Calibrated LR).* ***(B)*** *Weight Gain Paradox: higher weight_gain_pct* $\rightarrow$ *monotonically higher SHAP, contradicting* $r \approx -0.23$ *marginal correlation.*

The Weight Gain Paradox

Paradox

weight_gain_pct ranked **top-3 in feature importance** despite a **negative** linear correlation with *st_pres* ($r \approx -0.23$)

Statistical explanation:

- Linear correlation = marginal association only
- Tree/SHAP models capture **non-linear + interaction effects**
- Weight gain $\times$ strain $\times$ lipid profile $\Rightarrow$ context-dependent risk

Biological hypothesis:

- Strain-specific *bile acid signalling* (FXR/FGF19 axis)
- *Metabolic efficiency* vs. *lithogenic predisposition*
- **Multi-hit model**: weight gain modulates risk only in combination with strain + lipid context

Strain-Level Error Analysis

Where Does the Model Fail?

Strain	n	Mistakes	Error Rate	FN Rate
BKSChpLt.HRS-Cpe⟨fat⟩/J	35	13	37.1%	42.1%
B6(Cg)-Tub⟨tub⟩/J	49	11	22.4%	50.0%
C57BL/6J	36	10	27.8%	46.2%
B6.Cg-A⟨y⟩/J	52	4	7.7%	20.0%
B6.Cg-Lep⟨ob⟩/J	38	2	5.3%	50.0%

Key Finding

BKSChpLt.HRS-Cpe⟨fat⟩/J: 37.1% error rate and 42.1% FN rate among Stone-positive animals ⇒ biological/measurement heterogeneity – priority target for follow-up

Discussion & Conclusion

Limitations

- **Single cohort** (291 animals) – no external validation
- **Class imbalance**: 62 Stone vs. 229 No-Stone limits stability of estimates
- **Single imputation** (median) on high-missingness columns
  $\rightarrow$ artificially shrinks variance

Conclusion

- End-to-end ML pipeline for gallstone prediction in obesity-prone murine models
- **Calibrated LR**: Recall 0.771 – best for screening (minimise missed cases)
- Full pipeline **outperforms** univariate baseline (Recall +0.16, Precision +0.33)
- **Weight Gain Paradox**: non-linear predictors invisible to univariate analysis
- SHAP + strain error analysis $\Rightarrow$ transparent, biologically interpretable model
- Confusion matrix: 37 TP, 25 FN – room for improvement in Stone-positive detection

Future Work

Longitudinal features · Bayesian calibration · Genetic/transcriptomic covariates · Multi-cohort validation · MICE imputation

Thank You

Questions?